
3AL Tutorial 1 2024 Solutions

1. (a) Prove that $Z(D_4) = \{1, r^2\}$.
- (b) Determine the order of each element in D_4 .

(a) By the definition of identity element, 1 commutes with all elements of D_4 . Thus $1 \in Z(D_4)$. Now we show that $r^2 \in D_4$. That is, $r^2a = ar^2$ for all $a \in D_4$. Recall that $D_4 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ where $r^4 = s^2 = 1$ and $rsr = s$. We have

$$\begin{aligned}
 1r^2 &= r^2 = r^21 \\
 rr^2 &= r^3 = r^2r \\
 r^3r^2 &= r^5 = r^2r^3 \\
 sr^2 &= (rsr)r^2 = rsr^3 = r(rsr)r^3 = r^2s \\
 (sr)r^2 &= (rsr)r^3 = rs = r(rsr) = r^2(sr) \\
 (sr^2)r^2 &= s = rsr = r(rsr)r = r^2(sr^2) \\
 (sr^3)r^2 &= sr = (rsr)r = r(rsr)r^2 = r^2(sr^3).
 \end{aligned}$$

Thus, $r^2 \in Z(D_4)$. We need to now show that every other element fails to be in $Z(D_4)$. Notice the following:

$$\begin{aligned}
 rs &= r(rsr) = r^2(rsr)r = r^3(rsr)r^2 = sr^3 \neq sr \implies r, s \notin Z(D_4) \\
 r^3s &= r^3(rsr) = sr \neq sr^3 \implies r^3 \notin Z(D_4) \\
 r(sr) &= s \neq sr^2 = (sr)r \implies sr \notin Z(D_4) \\
 r(sr^2) &= sr \neq sr^3 = (sr^2)r \implies sr^2 \notin Z(D_4) \\
 r(sr^3) &= sr^2 \neq s = (sr^3)r \implies sr^3 \notin Z(D_4).
 \end{aligned}$$

Thus, $Z(D_4) = \{1, r^2\}$.

(b) $|1| = 1$, $|r| = |r^3| = 4$, $|s| = |sr| = |sr^2| = |sr^3| = |r^2| = 2$.

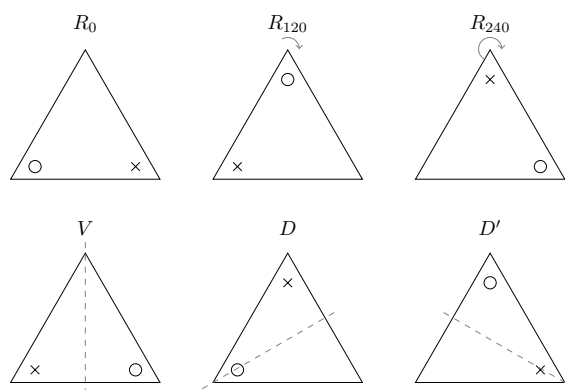
The method for finding the orders is just take the element and multiply by itself until you get the identity. For example,

$$(sr)(sr) = s(rsr) = ss = 1 \implies |sr| = 2.$$

2. (a) Draw the symmetries of an equilateral triangle. (Something similar to the diagram in Example 1.7 of the notes)
- (b) Find a group isomorphic to D_3 .

NOTE: You don't have to write a full proof that the two groups are isomorphic. Just write a convincing statement of why a certain group is isomorphic to D_3 .

(a)



- (b) Consider the groups S_3 and D_3 . We can represent them by

$$S_3 = \{\varepsilon, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\},$$

where $\sigma^3 = \tau^2 = \varepsilon$ and $\sigma\tau\sigma = \tau$, and

$$D_3 = \{1, r, r^2, s, sr, sr^2\}$$

where $r^3 = s^2 = 1$ and $rsr = s$. Relabelling r as σ and s as τ shows us that these two groups are “the same”.

3. Let $\varphi : G \rightarrow G'$ be a homomorphism. Prove the following:

- (a) $\ker \varphi$ is a subgroup of G ,
- (b) φ is a monomorphism if and only if $\ker \varphi = \{1_G\}$,
- (c) $\text{im } \varphi$ is a subgroup of G' ,
- (d) φ is an epimorphism if and only if $\text{im } \varphi = G'$.

(a) We know that $\varphi(1_G) = 1_{G'}$, thus $1_G \in \ker \varphi$. Let $a, b \in \ker \varphi$. We need to show $ab^{-1} \in \ker \varphi$. That is, $\varphi(ab^{-1}) = 1_{G'}$. We have

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_{G'}1_{G'}^{-1} = 1_{G'}.$$

Thus, $\ker \varphi \leq G$.

(b) Suppose φ is a monomorphism. We know that $\varphi(1_G) = 1_{G'}$ from homomorphism properties and so $1_G \in \ker \varphi$. Since φ is injective, there is only one element that maps to $1_{G'}$. Thus, $\ker \varphi = \{1_G\}$.

Now suppose that $\ker \varphi = \{1_G\}$. Let $a, b \in G$ with $\varphi(a) = \varphi(b)$. Then $1_{G'} = \varphi(a)\varphi(b)^{-1} = \varphi(ab^{-1})$. Then $ab^{-1} \in \ker \varphi$. Thus $ab^{-1} = 1_G$ and so $a = b$. Hence φ is a monomorphism.

(c) We have $\varphi(1_G) = 1_{G'}$, thus $1_{G'} \in \text{im } \varphi$. Let $a, b \in \text{im } \varphi$. Then there exist $c, d \in G$ such that $\varphi(c) = a$ and $\varphi(d) = b$. We need to show that $ab^{-1} \in \text{im } \varphi$. Note that

$$ab^{-1} = \varphi(c)\varphi(d)^{-1} = \varphi(cd^{-1}),$$

where $cd^{-1} \in G$. Thus, $ab^{-1} \in \text{im } \varphi$. Thus $\text{im } \varphi \leq G'$.

(d) Suppose φ is an epimorphism. Then for all $a \in G'$, there exists $b \in G$ such that $\varphi(b) = a$. Thus, $a \in \text{im } \varphi$ and so $G' \subseteq \text{im } \varphi$. But we already have $\text{im } \varphi \subseteq G'$. Thus $\text{im } \varphi = G'$.

Suppose that $\text{im } \varphi = G'$. Then for all $a \in G'$, there exists $b \in G$ such that $\varphi(b) = a$. That is, φ is an epimorphism.

4. Let G be a cyclic group. Consider the map $\varphi : G \rightarrow G$ defined by $\varphi(a) = a^2$ for all $a \in G$.

(a) Prove that φ is a homomorphism.

(b) Determine the possible orders of elements in $\ker \varphi$.

(a) We need to show that $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$. That is, $(ab)^2 = a^2b^2$. Since G is cyclic, we can write $G = \langle g \rangle$. Thus we have $a = g^{n_1}$ and $b = g^{n_2}$ for some $n_1, n_2 \in \mathbb{Z}$. Then,

$$(ab)^2 = (g^{n_1}g^{n_2})^2 = g^{2(n_1+n_2)} = g^{2n_1}g^{2n_2} = (g^{n_1})^2(g^{n_2})^2 = a^2b^2.$$

Thus, φ is a homomorphism.

(b) We have $a \in \ker \varphi$ if and only if $\varphi(a) = 1_G$. That is, $a^2 = 1_G$. So, each element in $\ker \varphi$ has order at most 2. So, the possible orders are 1 and 2.

5. Given the following groups and subgroups, list all of the distinct left cosets of the given subgroups (include a list of all elements in each coset).

(a) $G = \mathbb{Z}_{21}$ with $H = \langle [3] \rangle$,

(b) $G = \langle a \rangle$, where $|a| = 20$, with $H = \langle a^5 \rangle$.

(a) To ease notation, I will omit the equivalence class notation $[\cdot]$ in the solution. You're allowed to do this too in your answer, just state that you are doing it so it is clear that you understand that the letter n represents the equivalence class $[n]$.

$$H = \langle 3 \rangle = \{0, 3, 6, 9, 12, 15, 18\}$$

$$1 + H = \{1, 4, 7, 10, 13, 16, 19\}$$

$$2 + H = \{2, 5, 8, 11, 14, 17, 20\}$$

$$3 + H = \{3, 6, 9, 12, 15, 18, 0\} = H.$$

Notice that $3 + H = H$, $4 + H = 1 + H$, and so on. Therefore the distinct cosets are H , $1 + H$ and $2 + H$.

(b) Since $G = \langle a \rangle$, all elements of G have the form a^n for some $0 \leq n < 20$. So all

cosets will have the form $a^n H$.

$$H = \langle a^5 \rangle = \{1, a^5, a^{10}, a^{15}\}$$

$$aH = \{a, a^6, a^{11}, a^{16}\}$$

$$a^2 H = \{a^2, a^7, a^{12}, a^{17}\}$$

$$a^3 H = \{a^3, a^8, a^{13}, a^{18}\}$$

$$a^4 H = \{a^4, a^9, a^{14}, a^{19}\}$$

$$a^5 H = \{a^5, a^{10}, a^{15}, 1\} = H.$$

Since the cosets $a^n H$ start repeating when n reaches 5, all the preceding cosets are the distinct ones. So the distinct cosets are $H, aH, a^2 H, a^3 H$, and $a^4 H$.

6. Consider the (additive) group $\mathbb{Q}$ with subgroup $\mathbb{Z}$.

(a) Show that all cosets of $\mathbb{Z}$ of the following form are distinct, where $n \in \mathbb{N}^+$,

$$\frac{1}{n} + \mathbb{Z} = \left\{ \frac{1}{n} + z : z \in \mathbb{Z} \right\}.$$

NOTE: Distinct means if $n \neq m$, then $\frac{1}{n} + \mathbb{Z} \neq \frac{1}{m} + \mathbb{Z}$.

(b) Prove or disprove: $|\mathbb{Q} : \mathbb{Z}|$ is finite.

(a) We use proof by contradiction. Suppose that two such cosets are equal. That is, $\frac{1}{n} + \mathbb{Z} = \frac{1}{m} + \mathbb{Z}$ for some distinct $n, m \in \mathbb{N}^+$. Without loss of generality, assume that $n > m$. By the properties of cosets, this means that $-\frac{1}{n} + \frac{1}{m} = \frac{n-m}{nm} \in \mathbb{Z}$. Since this quotient is an integer, we must have $nm \mid (n-m)$. However,

$$n - m < n \leq nm.$$

This means we must have $n - m = 0$. This is a contradiction since we assume n and m are distinct.

NOTE: The first inequality, $n - m < n$, holds since $n > m$ and $m \in \mathbb{N}^+$. The second inequality, $n \leq nm$, holds since $m \in \mathbb{N}^+$ and so $m \geq 1$.

Alternatively, we can say that since $0 < \frac{1}{n}, \frac{1}{m} \leq 1$, it is impossible for $-\frac{1}{n} + \frac{1}{m}$ to be an integer other than 0. This can only hold if $n = m$, a contradiction.

(b) There are infinitely many cosets of the form $\frac{1}{n} + \mathbb{Z}$ thus $|G : \mathbb{Z}|$ is infinite.